Lecture 13 recap

In Lecture 13:

- Went through examples on using LTs to solve IVPs with discontinuous forcing

Lecture 14

In Lecture 14, we will:

1. Introduce the Dirac delta and use it to model impulsive forcing
2. Introduce convolutions as a tool to calculate ILTs

Impulses

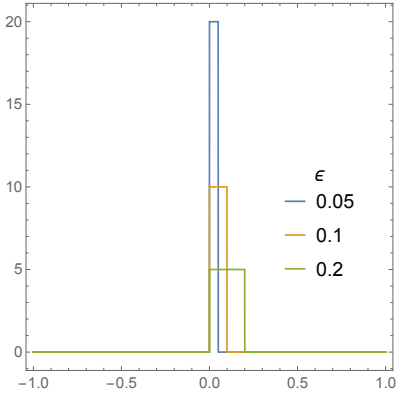
Consider the rectangular pulse of unit area:

$$\delta_\epsilon(t) = \begin{cases} 0 & t \leqslant 0 \\ \frac{1}{\epsilon} & 0 < t \leqslant \epsilon \\ 0 & t > \epsilon \end{cases}, \qquad \epsilon > 0$$

Useful for modelling when something that happens over a negligible time period, i.e., an impulse

The 'strength' of this impulse is

$$\int_0^\infty \delta_\epsilon(t)\,\mathrm{d}t = \int_0^\epsilon \frac{1}{\epsilon}\,\mathrm{d}t = 1$$



Limit as $\epsilon \to 0$

1. $\lim_{\epsilon \to 0} \delta_\epsilon = 0$ for all $t \neq 0$
2. $\lim_{\epsilon \to 0} \int_0^\infty \delta_\epsilon \, \mathrm{d}t = 1$
3. $\lim_{\epsilon \to 0} \mathscr{L}(\delta_\epsilon) = \lim_{\epsilon \to 0} \dfrac{1 - \mathrm{e}^{-\epsilon s}}{\epsilon s} = 1$ (L'Hôpital's rule)

The Dirac delta, $\delta(t)$

Definition

We define $\delta(t)$ such that

1. $\delta(t) = 0$ for all $t \neq 0$,
2. $\displaystyle\int_{-\infty}^{\infty} \delta(t)\mathrm{d}t = \int_{0}^{a} \delta(t)\mathrm{d}t = 1$ for all $a > 0$, and
3. $\mathscr{L}(\delta) = 1$.

Sifting property

- Since $\delta(t)$ is zero for $t \neq 0$, it 'sifts out' the value of a function at $t = 0$

$$\int_0^\infty f(t)\delta(t)\mathrm{d}t = f(0)$$

- Let $c \geqslant 0$, then

1. $\displaystyle\int_0^\infty f(t)\delta(t-c)\ \mathrm{d}t = f(c)$

2. $\displaystyle\mathscr{L}\Big(\delta(t-c)\Big) = \int_0^\infty \mathrm{e}^{-st}\delta(t-c)\ \mathrm{d}t = \mathrm{e}^{-sc}$

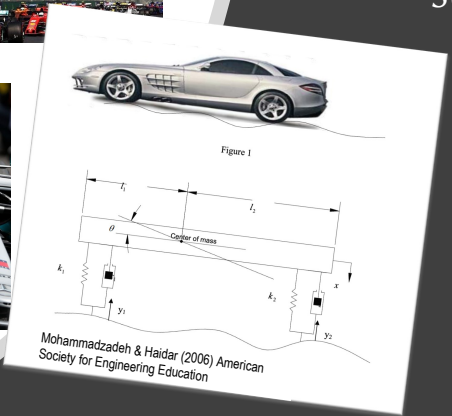
Exercise 3.19: Solve $y'(t) - y(t) = \delta(t-1)$ where $y(0) = 0$

We solve the problem using Laplace transforms.

Case study 4: Formula 1 car suspension

https://www.planetf1.com/news/sydney-australian-grand-prix-bid/

https://www.mercedesamgf1.com/en/news/2019/06/the-suspension-of-a-formula-one-car/



- Read all about formula 1 car suspensions here https://www.mercedesamgf1.com/en/news/2019/06/the-suspension-of-a-formula-one-car/
- **Mathematical models lead to ODEs which can be solved using Laplace transforms.**

Convolution

Definition (Convolution)

Given two functions, f and g, defined on $[0, \infty)$, their convolution, $f * g$ is defined

$$f * g = \int_0^t f(t-\tau)g(\tau)\mathrm{d}\tau$$

Basic properties

- Convolutions are commutative, i.e., $f * g = g * f$
- In general, $1 * f \neq f$, as

$$(1 * f)(t) = \int_0^t f(\tau)\mathrm{d}\tau, \quad \text{i.e., an antiderivative}$$

Exercise 3.20: Calculate the convolution of e^t with e^{2t}

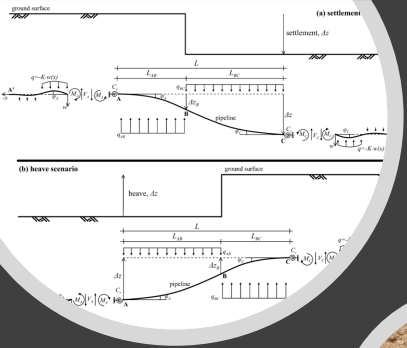
Exercise 3.21: Calculate $\mathscr{L}^{-1}\left\{\dfrac{1}{(s-1)(s-2)}\right\}$

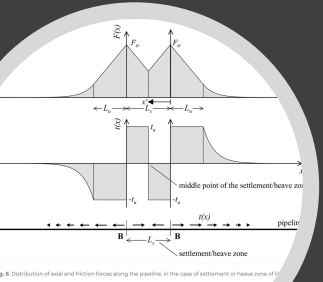
Laplace transform of the convolution

Let f and g be functions with Laplace transforms, then

$$\mathscr{L}(f * g) = F(s)G(s)$$

Laplace transforms convert convolutions into multiplication





g. 6. Distribution of axial and friction forces along the pipeline, in the case of settlement or heave zone of l

Case study 5: Critical pipeline performance

- Buried pipelines are critical lifeline systems, supplying drinking water and natural gas to communities. Damage to these pipelines caused by ground deformation and related hazards like earthquakes and landslides can lead to severe socioeconomic losses to unprepared communities.
- Mitigation measures employ mathematical models of these structures and these often involve ordinary differential equations with discontinuous forcing and impulsive functions
- Images from Lee et al (2016) Quarterly Journal of Engineering Geology and Hydrogeology, and Kouretzis et al (2015) Canadian Geotechnical Journal

Exercise 3.22: Solve the following integro-differential equation

$$y'(t) + 4\int_0^t y(\tau)\mathrm{d}\tau = 0 \qquad \text{where} \qquad y(0) = 3$$

Laplace transform of an antiderivative

$$\mathscr{L}\left(\int_0^t f(\tau)\mathrm{d}\tau\right) = \mathscr{L}(1 * f) = \mathscr{L}(1)F(s) = \frac{F(s)}{s}$$